HACKMERIDIAN LISBON · GENESIS (IDEA -> MVP) · STELLAR UNLOCKED TRACK

AuditLedger - Executive Team Brief

Problem-Solution-Market Fit, Architecture Advantages & 2-Minute Live Demo Script

1. PROBLEM-SOLUTION & MARKET FIT (100% ALIGNED)

DIMENSION	WHY IT SOLVES A REAL B2B PAIN POINT & WINS JUDGES
1. The Real Problem (B2B Pain Point)	Every day, CFOs, finance teams, legal/IP departments, and companies send **CSV spreadsheets, Excel exports, and PDFs** (accounting ledgers, bank statements, patent portfolios, audit trails) by email to external auditors, banks, or investors. **Anyone along the chain can open a CSV or PDF and silently change a number or row in 5 seconds without leaving a trace.**
2. Why Current Solutions Fail	• **PDF Signatures (DocuSign):** Do not work for CSV/Excel data pipelines, and if a file is modified they only say *"Signature Invalid"* - they **cannot tell you which row out of 5,000 was altered**. • **Shared Cloud ERPs:** Force external auditors and clients onto the same expensive software and **violate privacy** by uploading confidential financial/IP records to third-party servers.
3. Our Solution (AuditLedger)	• **100% Local Browser Privacy (GDPR-Safe):** The actual file never leaves the user's computer. Only cryptographic fingerprints are computed locally in the browser. • **Row-by-Row Tamper X-Ray (Merkle Tree):** If someone alters even **1 cent or 1 character** on Row #4 (or Row #1,450), the receiver immediately sees the exact modified row highlighted in red with the original vs. tampered value. • **Excel Re-Save Resilience (Canonical Merkle Match):** If someone opens a CSV in Excel and re-saves it (changing line endings `\r\n` or delimiters `;`), a raw SHA-256 file hash breaks, whereas our canonical Merkle Root proves the underlying data and numbers are still 100% intact.
4. Why Stellar? (Soroban Fit)	A company sealing thousands of financial reports, invoices, or IP snapshots per month cannot pay $5 - $20 in gas fees or wait minutes for finality. With **Stellar Soroban**: • **~5-Second Deterministic Finality** on Stellar ledger. • **Constant O(1) Fee (~0.00001 XLM):** Whether a CSV has **10 rows, 60 rows, or 10,000 rows**, our automatic browser Merkle Tree rolls all rows into **1 single 32-byte Master Digital Seal**. Only **1 Soroban smart contract call** is ever needed!

2. TECHNICAL COMPLETENESS & CAPABILITIES IMPLEMENTED

LIVE SOROBAN SMART CONTRACT (RUST)

- Deployed on Stellar Testnet (`CDF4JHYP...6VLKED`).
- **notarize_ledger:** Locks Document SHA-256, Merkle Root, and Totals Commitment on-chain.
- **attest_audit:** Multi-party External Auditor co-signature recorded on Stellar.
- **Protocol 23 Ready:** Direct JSON-RPC verification & Freighter wallet + 1-click Demo Key support.

UNIVERSAL DYNAMIC FILE ENGINE

- **Universal CSV Parser:** Auto-detects both `debit/credit` ERP exports AND single-column `date,description,amount` CSVs (mapping negative to Debit/Expense and positive to Credit/Inflow) plus any generic CSV, PDF, or JSON.
- **Unlimited Rows + Pagination:** Hashes all rows into the Merkle Tree with clean `25 / 50 / 100` table pagination and auto-jump to modified rows.

3. THE 2-MINUTE LIVE DEMO SCRIPT FOR JUDGES

1. **0:00 - 0:30 (The Problem):** *"When a company emails a financial CSV or IP report to an external auditor or investor, anyone can secretly edit a row in 5 seconds. And because of GDPR, you can never upload raw financial spreadsheets to a public blockchain."*
2. **0:30 - 1:15 (Role 1 - Sender Seals File on Stellar):** Upload any CSV or PDF in **1. Seal File (Sender)**. Point out that 100% of rows stay private in the browser. Click **Step 2: Lock & Seal on Stellar Now** (5 seconds, ~0.00001 XLM fixed fee whether 10 or 10,000 rows) and open the **Printable Seal Certificate** with the Stellar QR code.
3. **1:15 - 2:00 (Role 2 - Receiver Catches Tampering in Red):** Switch to **2. Verify File (Receiver)**. First show **1. Original File (100% Authentic)** + optional **External Auditor Co-Signature**. Then click **2. Simulate 1 Edited Row (Row #4)** (or upload the downloaded modified CSV): in 1 second, AuditLedger turns red, jumps straight to **Row #4**, and exposes the exact altered value!

Live App: https://lisbon-hackmeridian-production.up.railway.app

Soroban Contract: CDF4JHYPUJA67S62TVWFDLV7PDDGH4H7PGN4ANUUFLLCPZNCM76VLKED